



Sampling and Bias

Overview

Review of linear regression

Sampling and sampling bias

If there is time: Sampling distributions

Announcement

Homework 3 has been posted!

It is due on Gradescope on **Sunday September 27th at 11pm**

- **Be sure to mark each question on Gradescope!**

Jessica is going to have an R review session

- Time: Saturday 1-2pm
- Location: Bass L01-A

Also, keep attending the practice sessions for more practice!

Announcement: mini-exam

The mini-exam will **in class** on **Tuesday September 29th**

- Last 30 minutes of class

You should know/understand:

1. All the symbols we have used in class represent

- E.g., A pirate sampled $n = 20$ students and calculated the correlation between their grades on a mini-exam and how much they loved the sea (on a scale from 0 to 100). What symbol should the pirate use to denote this relationship?

2. Suppose we wanted to visualize the relationship mentioned above. How could we do this in R?

- E.g., What does this R code do? `plot(grades, sea_love)`

	Statistic	Parameter
Mean	$\bar{x}$	μ
Standard deviation	s	σ
Proportion	$\hat{p}$	π
Correlation	r	ρ
regression slope	b	β

A practice exam has been posted to Canvas

If you have accommodations for extra time, you will take the exam in Kline tower, 11th floor , room 1105B

Review of linear regression

Review: Regression

Regression is method of using one variable x to predict the value of a second variable y

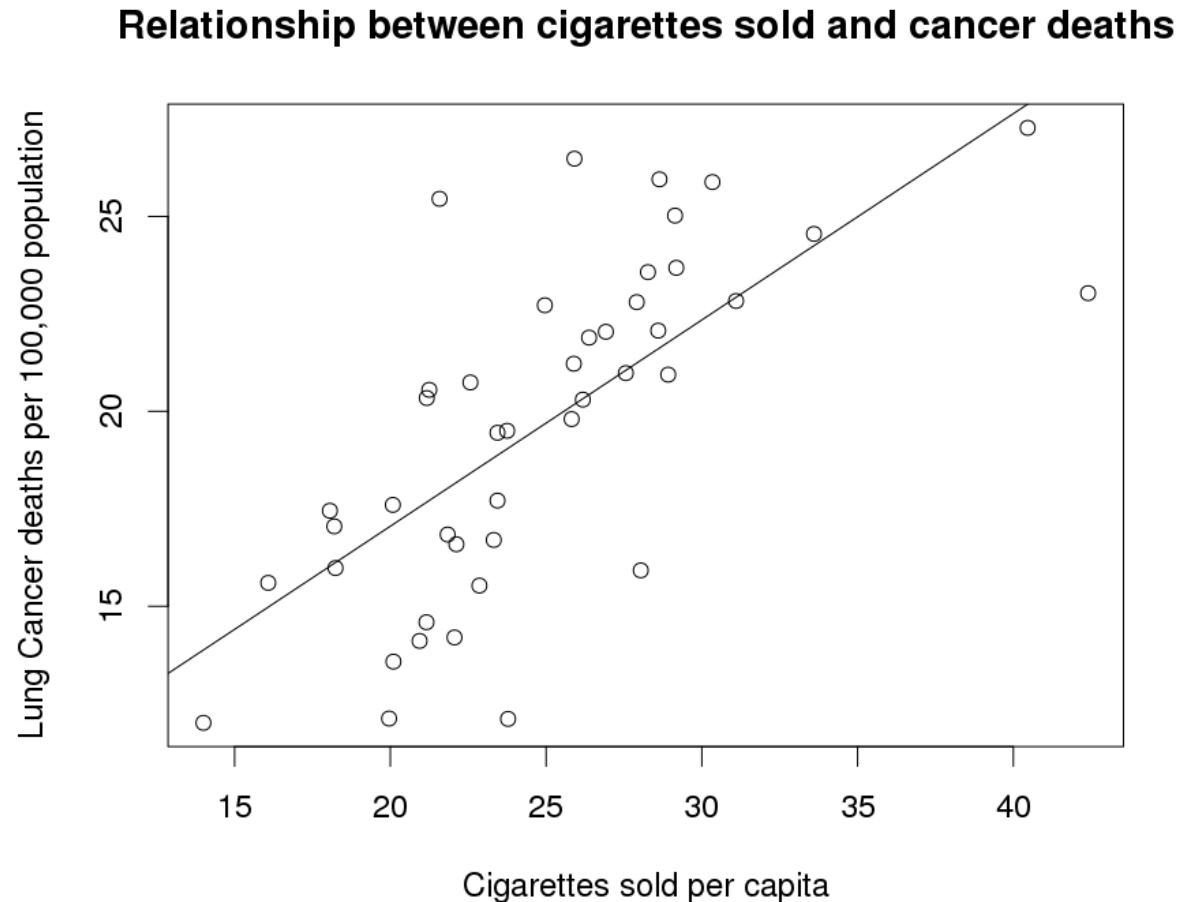
- i.e., $\hat{y} = f(x)$

In **linear regression** we fit a line to the data, called the **regression line**

$$\hat{y} = a + b \cdot x$$

$$\text{Response} = a + b \cdot \text{Explanatory}$$

Review: Cancer smoking regression line



$$\hat{y} = a + b \cdot x$$

R: `my_fit <- lm(y ~ x)`
`coef(my_fit)`

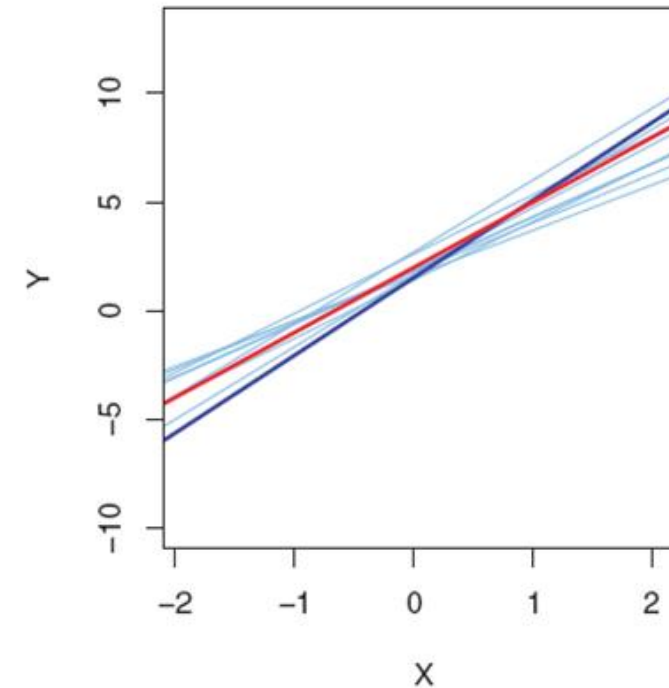
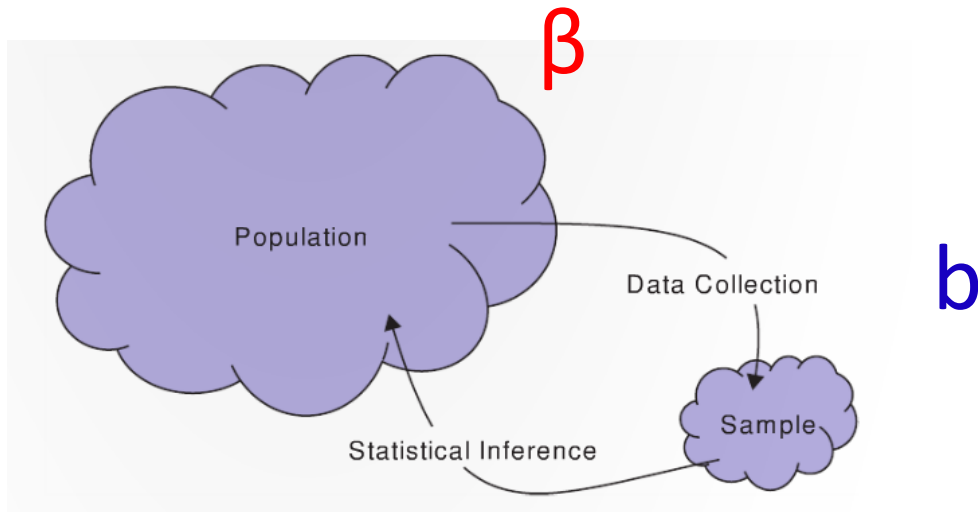
$$a = 6.47 \quad b = 0.0053$$

$$\hat{y} = 6.47 + .0053 \cdot x$$

Review: Notation

The Greek letter β is used to denote the slope of the **population**

The letter b is used to denote the slope of the **sample**

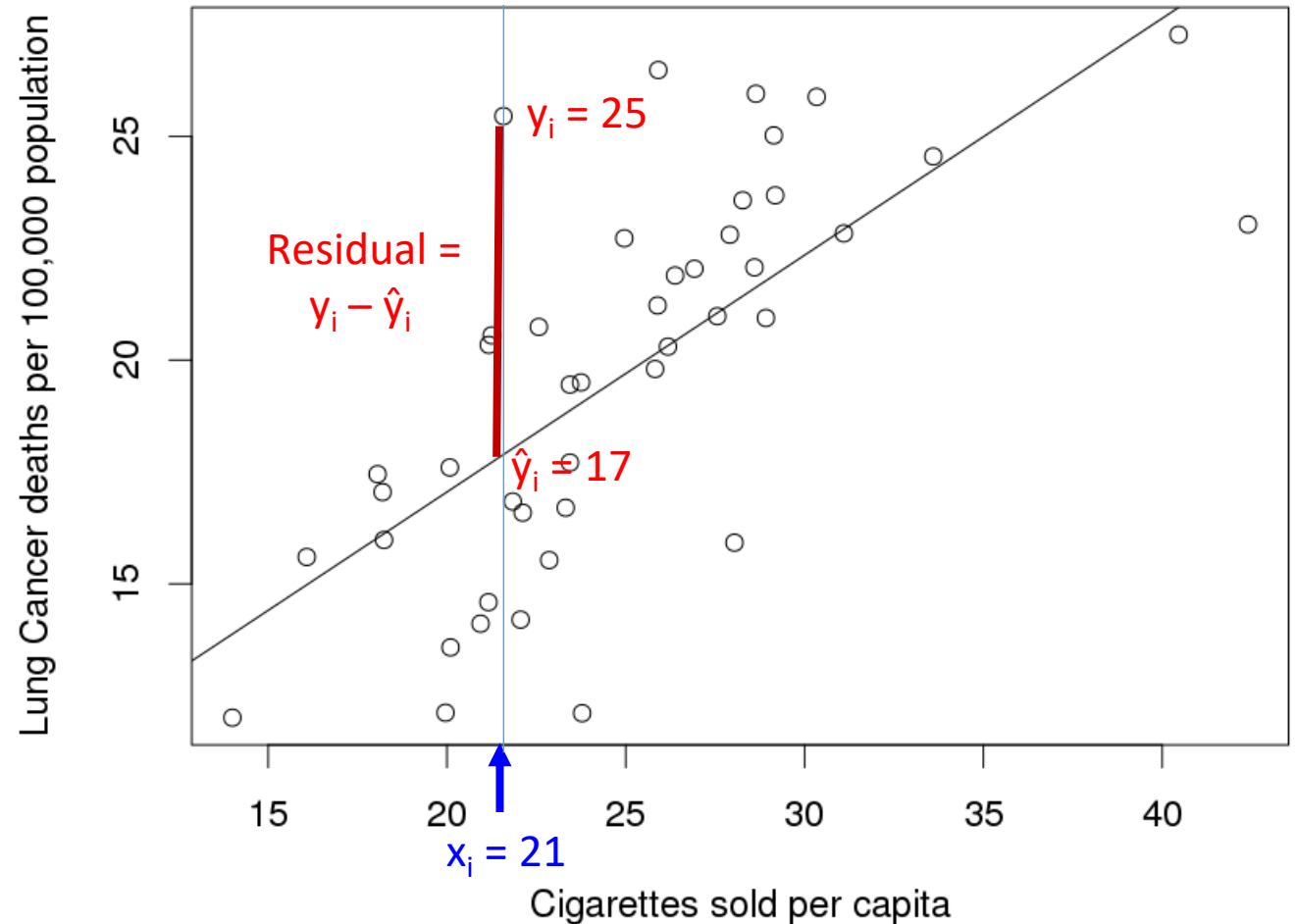


Review: residuals

The **residual** is the difference between an observed (y_i) and a predicted value ($\hat{y}_i$) of the response variable

$$\begin{aligned} \text{Residual}_i &= \\ \text{Observed}_i - \text{Predicted}_i &= \\ y_i - \hat{y}_i \end{aligned}$$

Relationship between cigarettes sold and cancer deaths



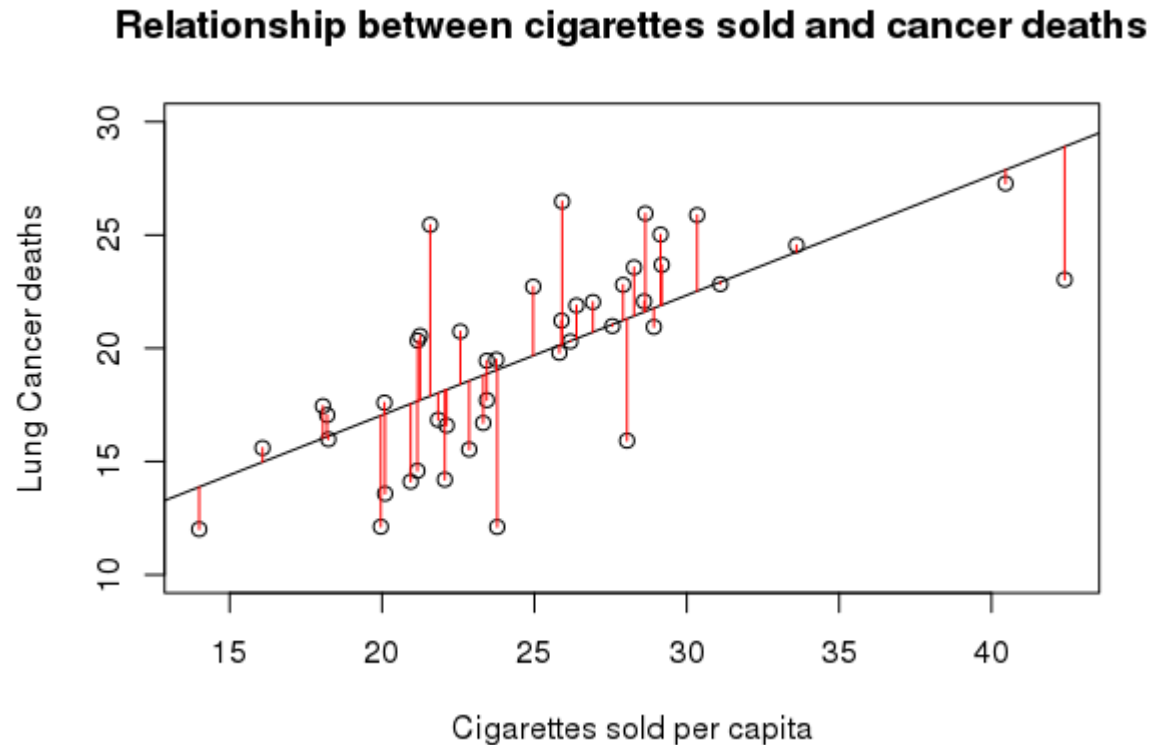
Review: Cancer smoking residuals

$$\hat{y} = 6.47 + 0.0053 \cdot x$$


Cig per Capita (x)	Cancer obs (y)	Cancer pred ($\hat{y}$)	Residuals (y - $\hat{y}$)
1,820	17.05	16.10	0.95
2,582	19.80	20.13	-0.33
1,824	15.98	16.12	-0.14
2,860	22.07	21.60	0.47
3,110	22.83	22.93	-0.10
3,360	24.55	24.25	0.30
40,460	27.27	27.88	-0.61

Review: Least squares line

The **least squares line**, also called '**the line of best fit**', is the line which minimizes the sum of squared residuals

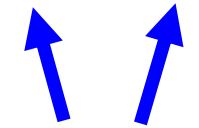


[Find the line of best fit](#)

Review: Cancer smoking residuals

Cancer obs (y)	Cancer pred ($\hat{y}$)	Residuals (y - $\hat{y}$)	Residuals ² (y - $\hat{y}$) ²
17.05	16.10	0.95	0.90
19.80	20.13	-0.33	0.11
15.98	16.12	-0.14	0.02
22.07	21.60	0.47	0.22
22.83	22.93	-0.10	0.01
24.55	24.25	0.30	0.09
27.27	27.88	-0.61	0.37
23.57	21.24	2.14	4.59

$$\hat{y} = a + b \cdot x$$



Find the a and b



That minimizes the sum
of the squared residuals

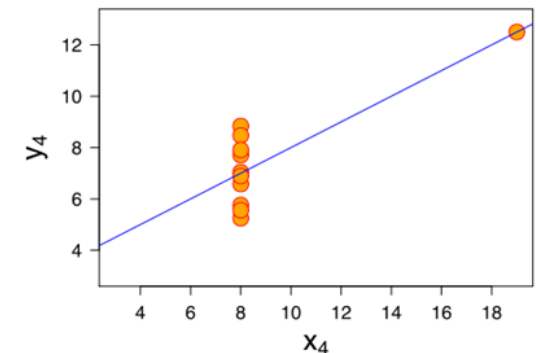
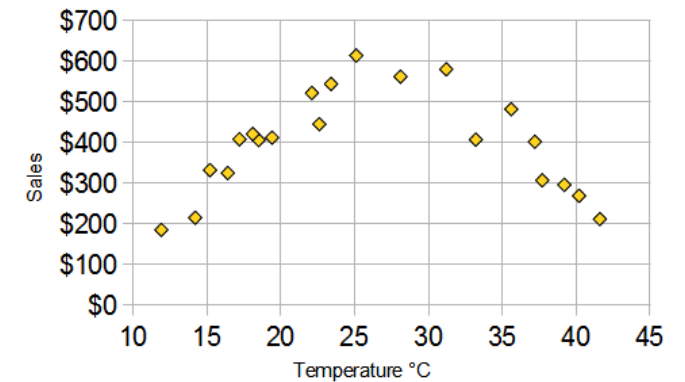
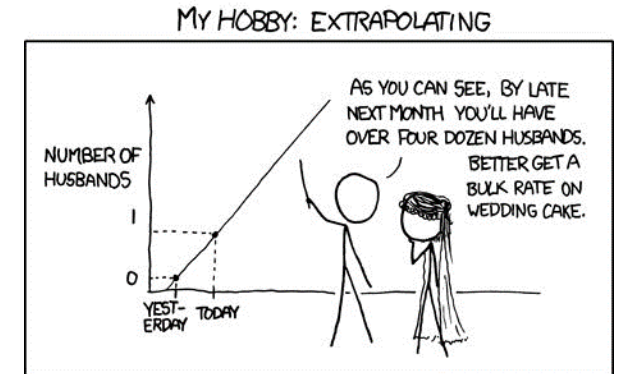
Review: Regression cautions

1. Avoid trying to apply the regression line to predict values far from those that were used to create the line

- i.e., do not extrapolate too far

2. Plot the data! Regression lines are only appropriate when there is a linear trend in the data

3. Be aware of outliers – they can have a huge effect on the regression line



Reivew: Regression lines in R

load the data

```
load("states_smoking.rda")
```

create a scatter plot and calculate the correlation

```
plot(smoking$CIG, smoking$LUNG)
```

Note: `plot(x, y)`



fit a regression model Note: `lm(y ~ x)`

```
lm_fit <- lm(smoking$LUNG ~ smoking$CIG)
```

get the a and b coefficients

```
coef(lm_fit)
```

add a regression line to the plot

```
abline(lm_fit)
```

The image features a dense field of three-dimensional question marks. Most of the question marks are a dark, matte grey color and are scattered across the frame, creating a textured background. In the center of the image, a single question mark stands out prominently, rendered in a bright, vibrant orange color. This orange question mark is slightly larger and more defined than the others. Overlaid on the orange question mark is the word "Questions?" in a clean, white, sans-serif font. The lighting is soft, casting gentle shadows and highlighting the three-dimensional nature of the symbols.

Questions?

Sampling

Where do samples/data come from?



Example: sampling 100 sprinkles



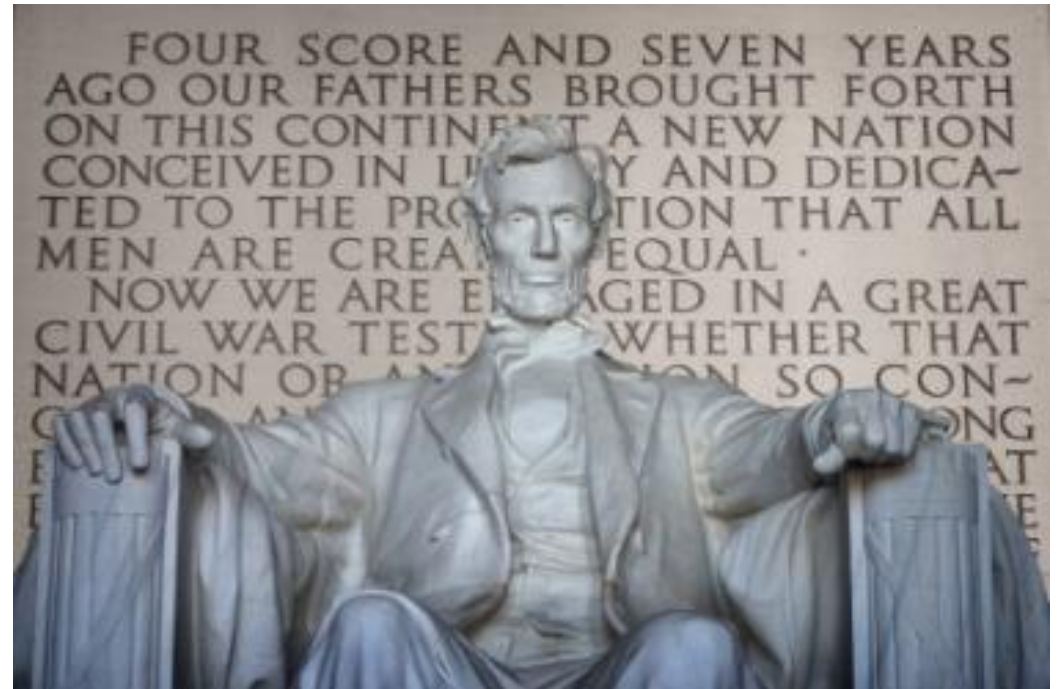
1	orange
2	red
3	green
4	white
5	white
6	white
7	white
8	white
9	red

The **sample size** (n) is the number of items in the sample
What is **n** in the sprinkle example?

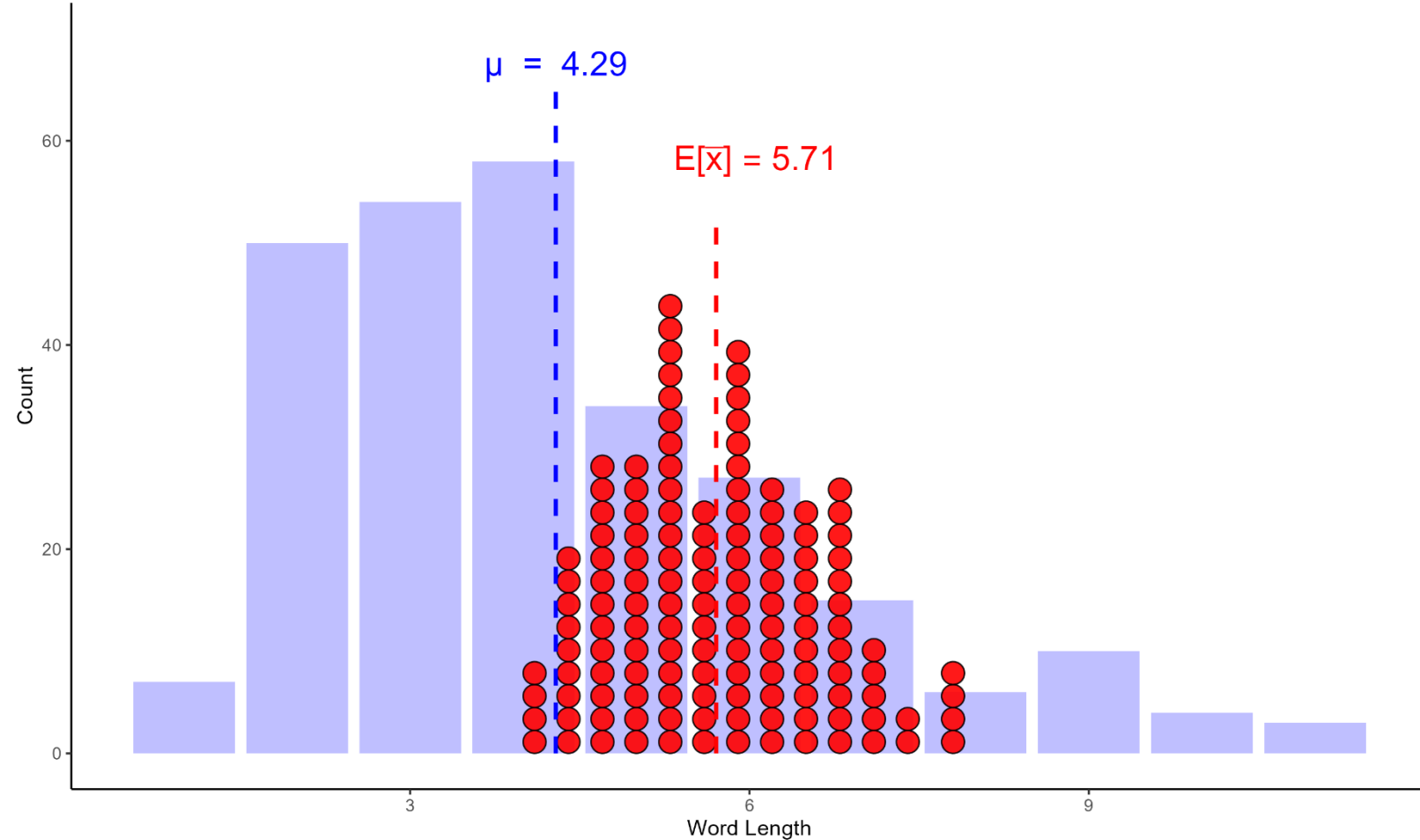
Sampling from the Gettysburg address

You filled out Gettysburg sampling worksheet survey on Canvas where you will randomly sample 10 words from the Gettysburg address

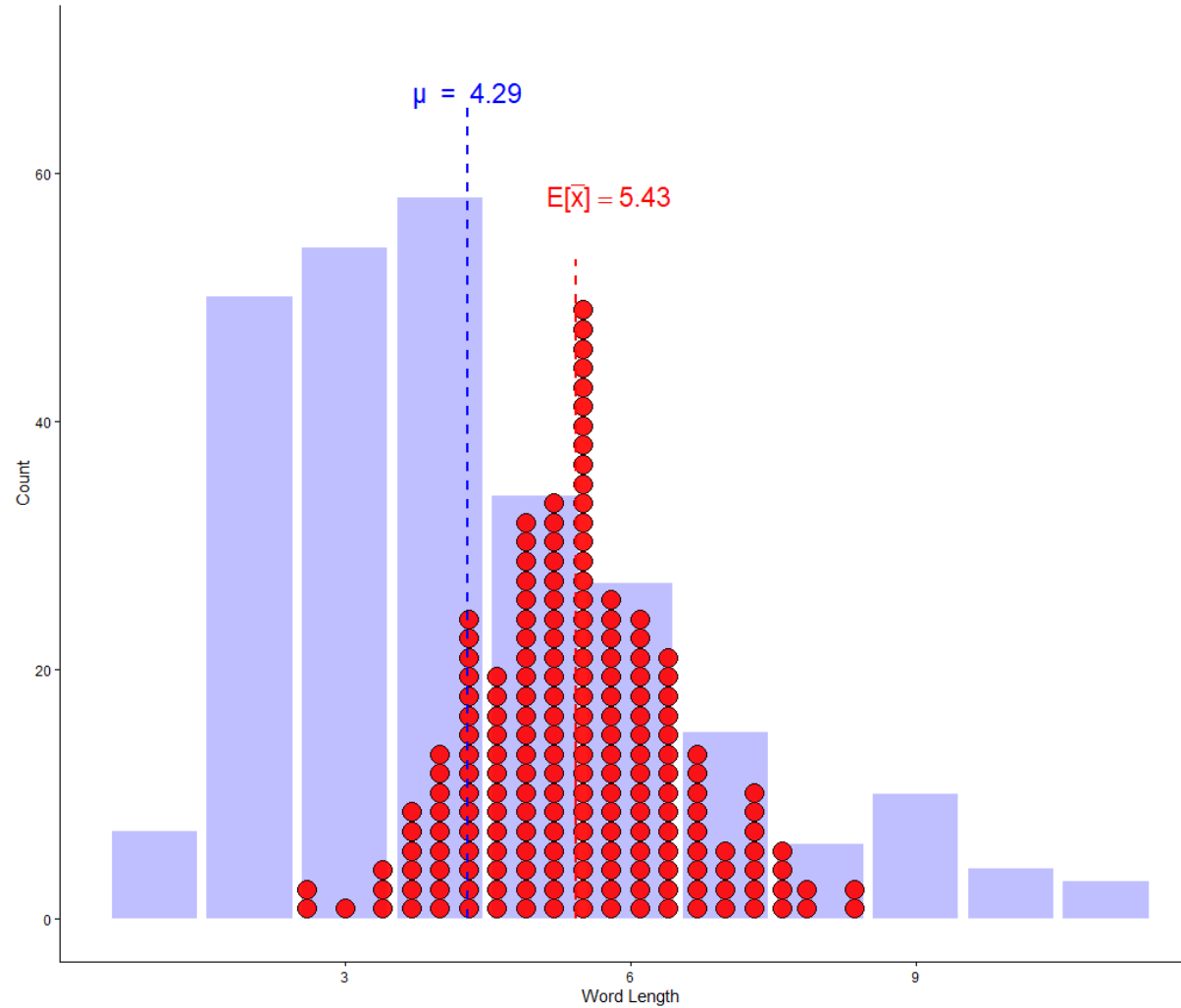
You reported the mean word $\bar{x}$ length of your 10 words



Bias and the Gettysburg address word length distribution



Your responses - fall 2026 (n = 200)

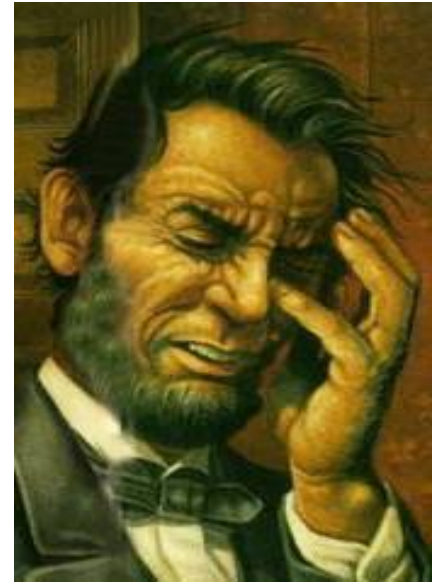
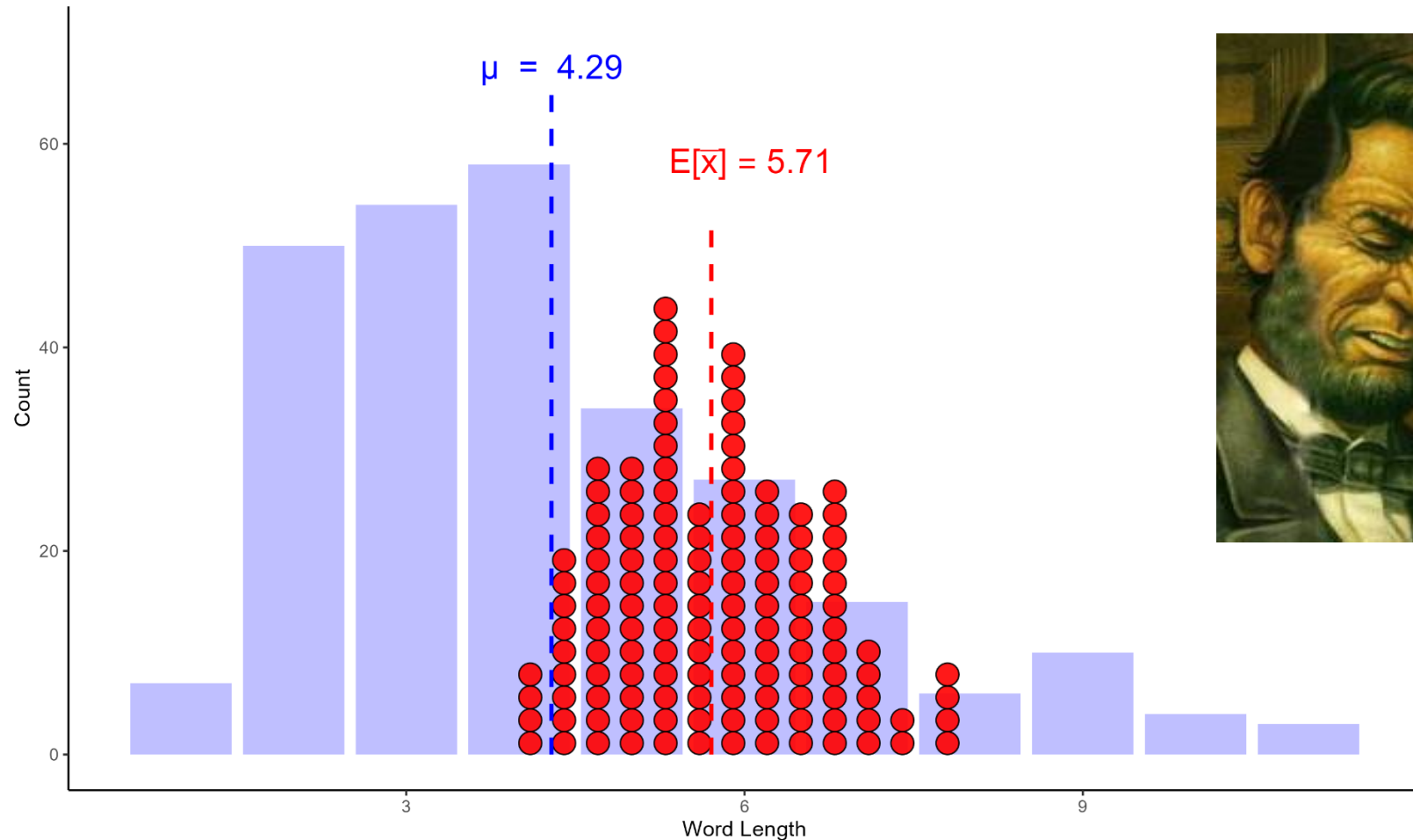


Bias and the Gettysburg address word length distribution

Bias is when the average statistic values does not equal the population parameter

Here:

$$E_s[\bar{x}] \neq \mu$$



Bias

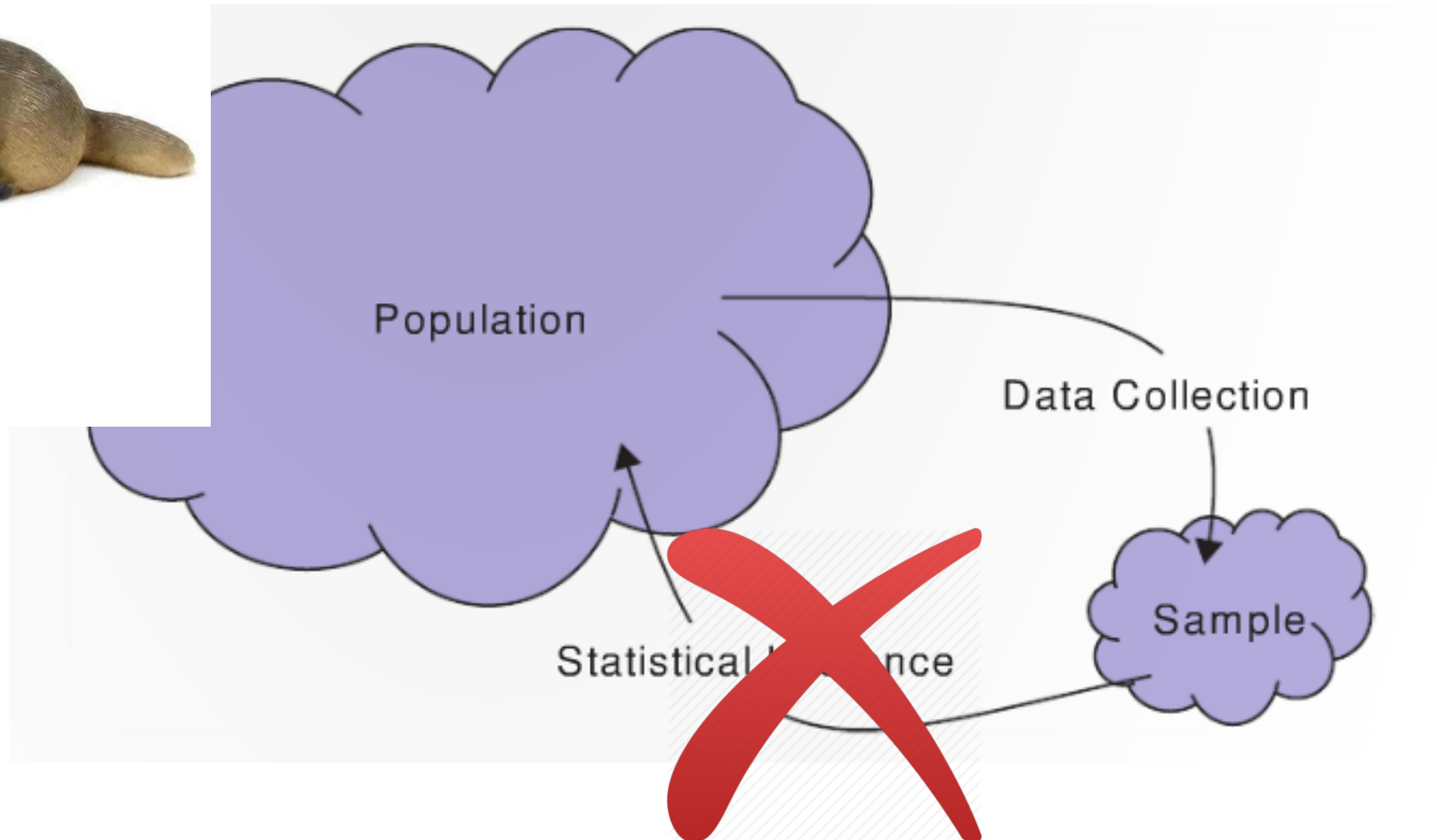
Sampling bias exists when the method of collecting the data causes the sample to inaccurately reflect the population

This leads to ***biased statistics*** where our average statistic value does not equal the parameter value

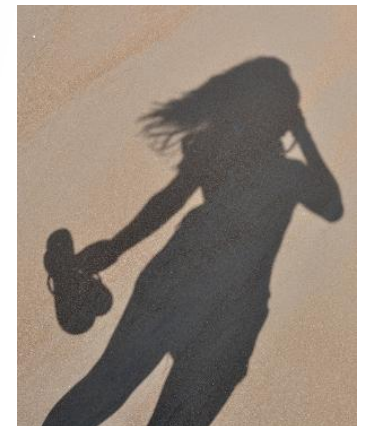
- E.g., $E_s[\bar{X}] \neq \mu$

Statistical bias

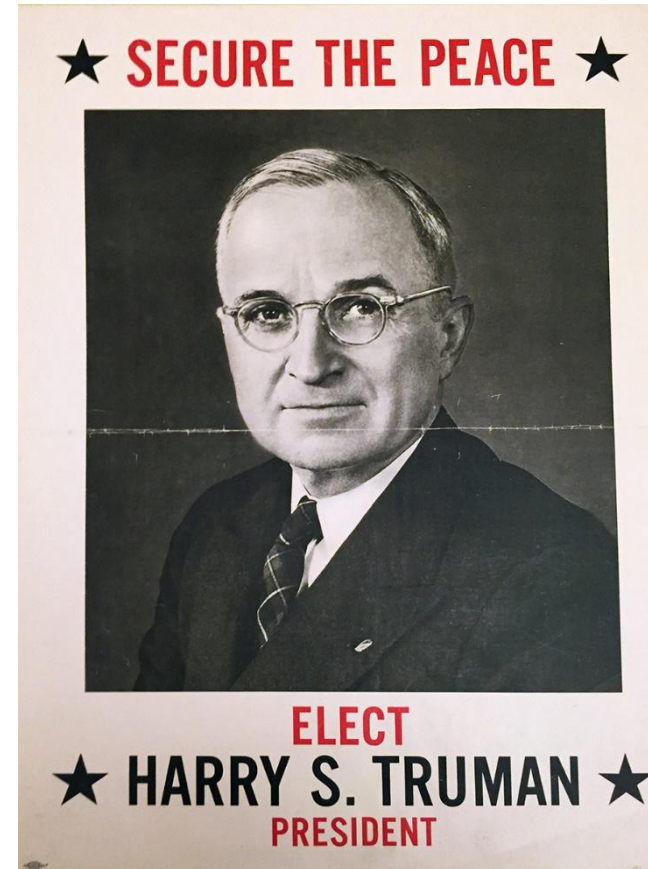
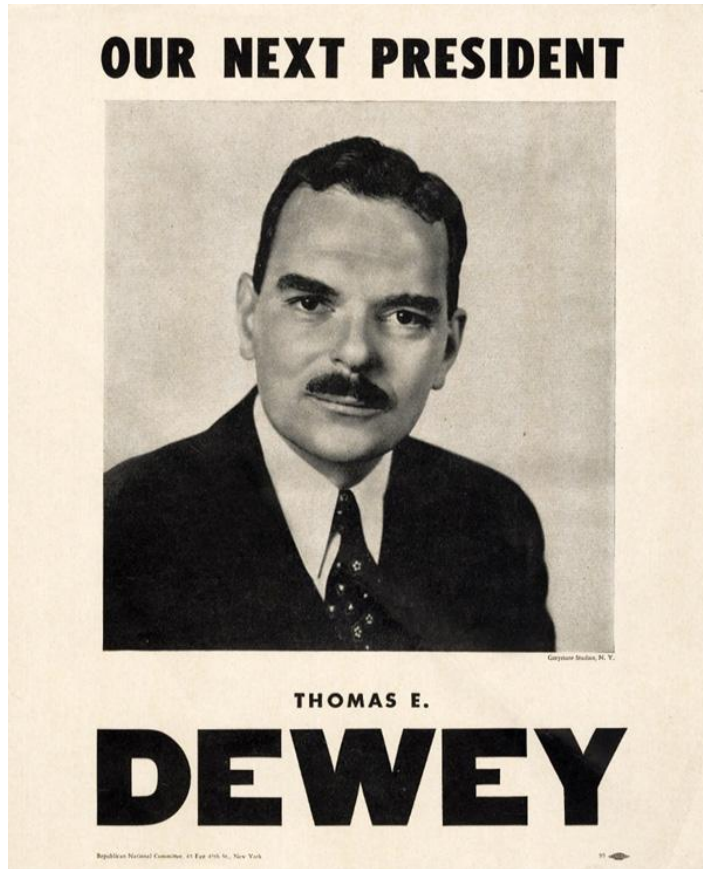
μ



$\bar{x}$



1948 US election



Newspaper title: Dewey Defeats Truman (1948)

The newspaper was published before the conclusion of the 1948 presidential election

The results were based on a large telephone poll which showed Dewey sweeping Truman

However, Harry S. Truman won the election

Q: What went wrong?



Basic questions for sampling

What is the population?

What is the sample?

Do they differ in a meaningful way?

To prevent bias: use simple random sample!

Simple random sample: each member in the population is equally likely to be in the sample

Allows for generalizations to the population!

Soup analogy



How do we select a random sample?

Mechanically:

- Flip coins

- Pull balls from well mixed bins

- Deal out shuffled cards, etc.

Use computer programs



Bias or no bias?

Practicalities...

It might not be feasible to randomly select equally from all members of a population

This might not be a problem as long as the sample is representative of the population

Example: If we wanted to know proportion of left-handed people in the US, randomly sampling Yale students might be sufficient

Need to think carefully to avoid bias!

Statistics requires thought!

Use your own reasoning:

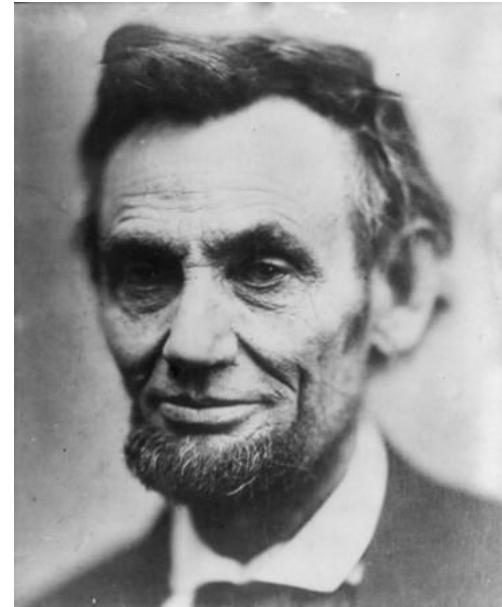
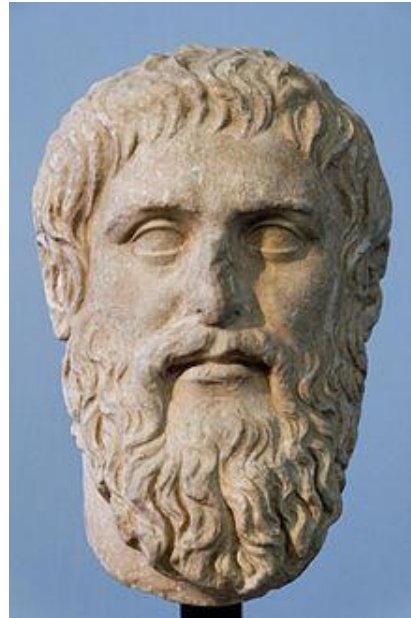
- What is the population I am interested in?

- Does the sample reflect the population of interest?

- Be your own worst critic!

From now on we are going to assume no bias!

Happy Plato and Lincoln

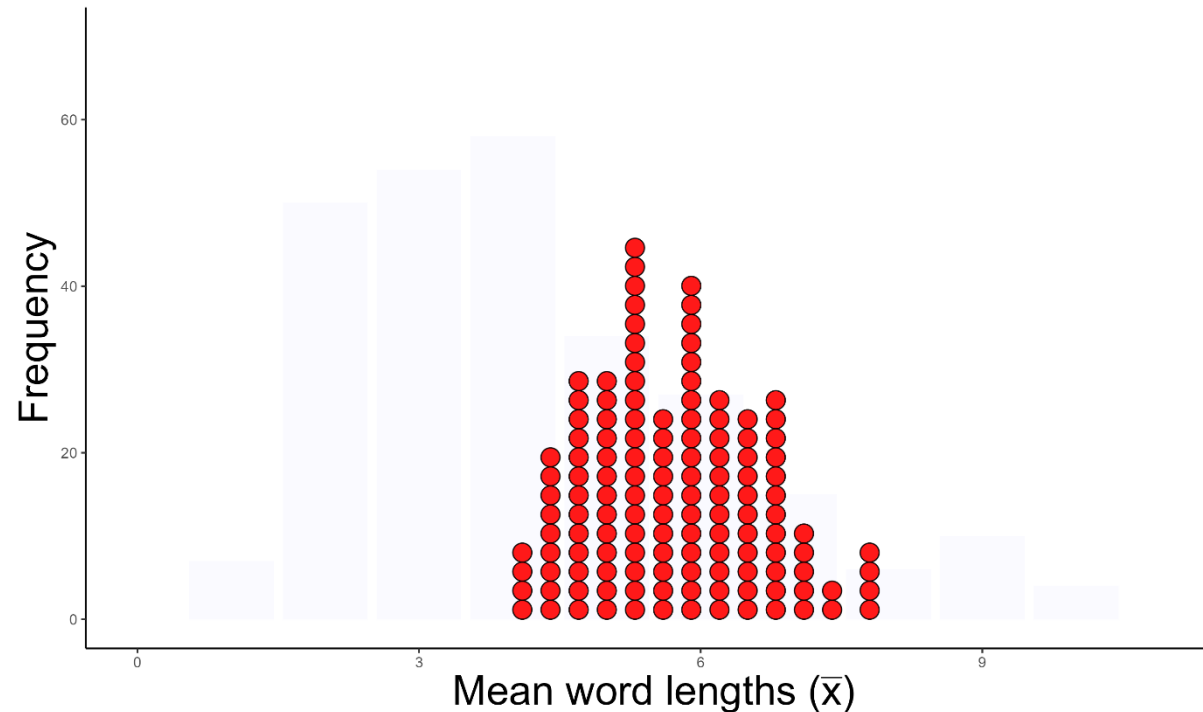


Our statistic values, on average, reflect the parameters

Sampling distributions

Recall for our distribution of Gettysburg word lengths...

Q: What does each dot that is plotted correspond to?



A: The mean length of 10 words ($\bar{x}$)

i.e., each point in our distribution is a **statistic**!

Sampling distribution

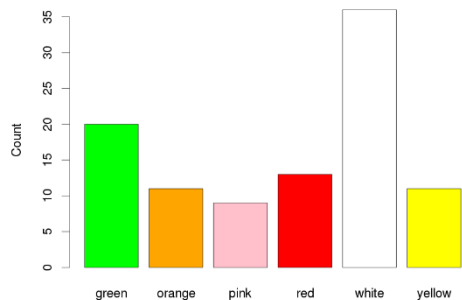
THEORETICAL CONCEPT

A **sampling distribution** is the distribution of sample statistics computed from different samples of the same size (n) from the same population

A sampling distribution shows us how the sample statistic varies from sample to sample

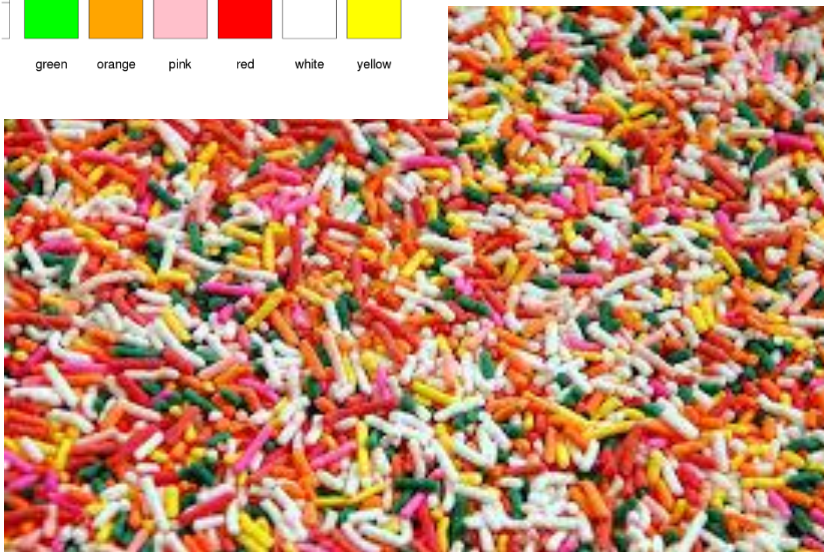
Note in general sampling distributions are a **theoretical concept** and not something that is actually created in real studies

- Although we will create a few of them as R stimulations so we can understand what this “theoretical” distribution is

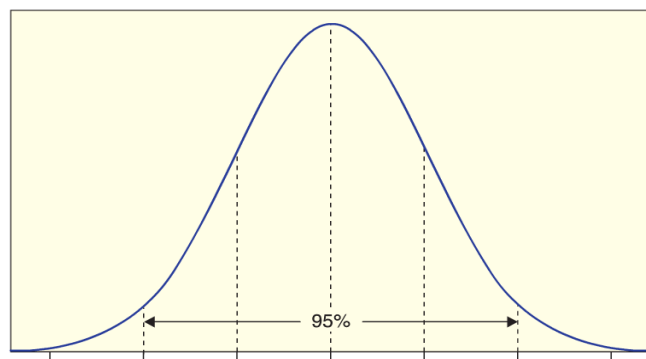


π_{red}

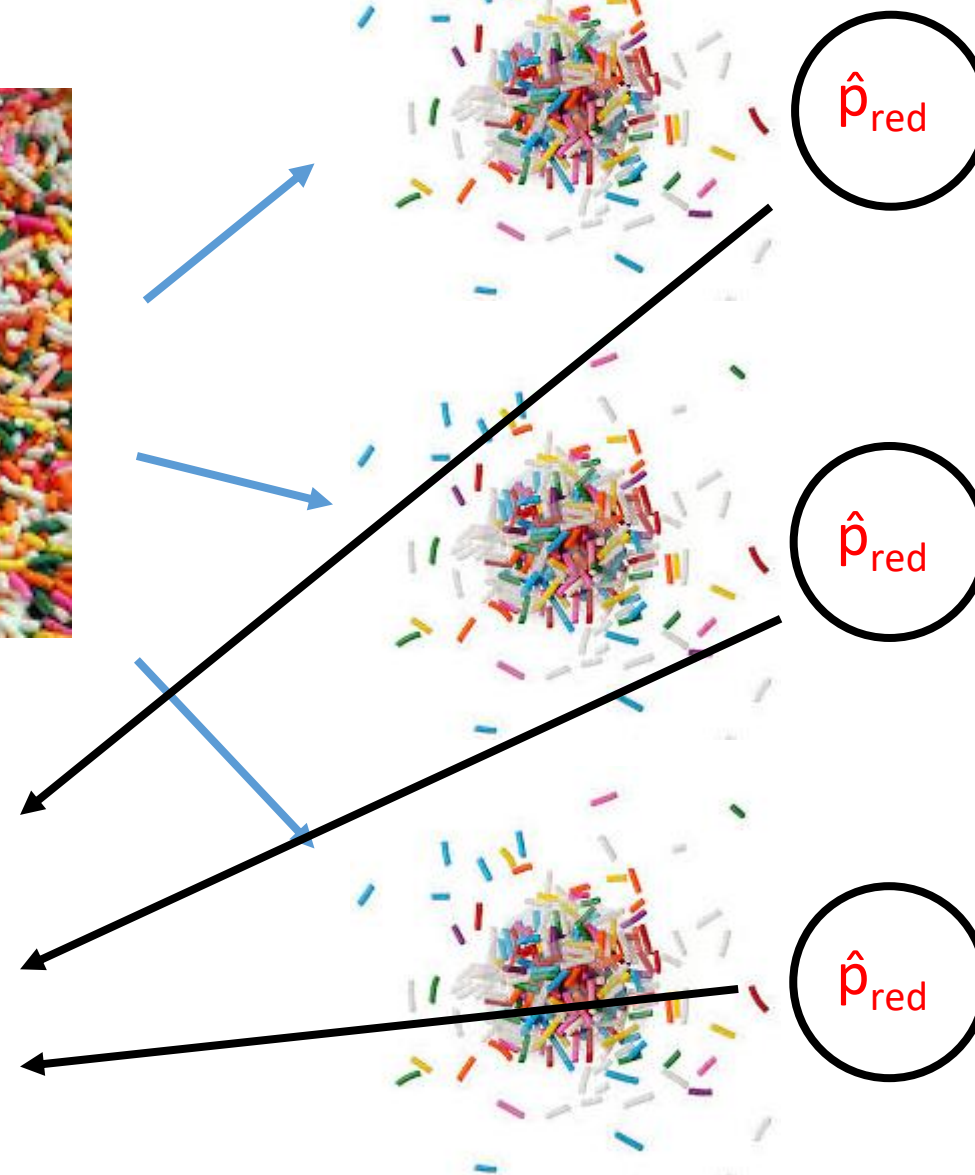
$n = 100$



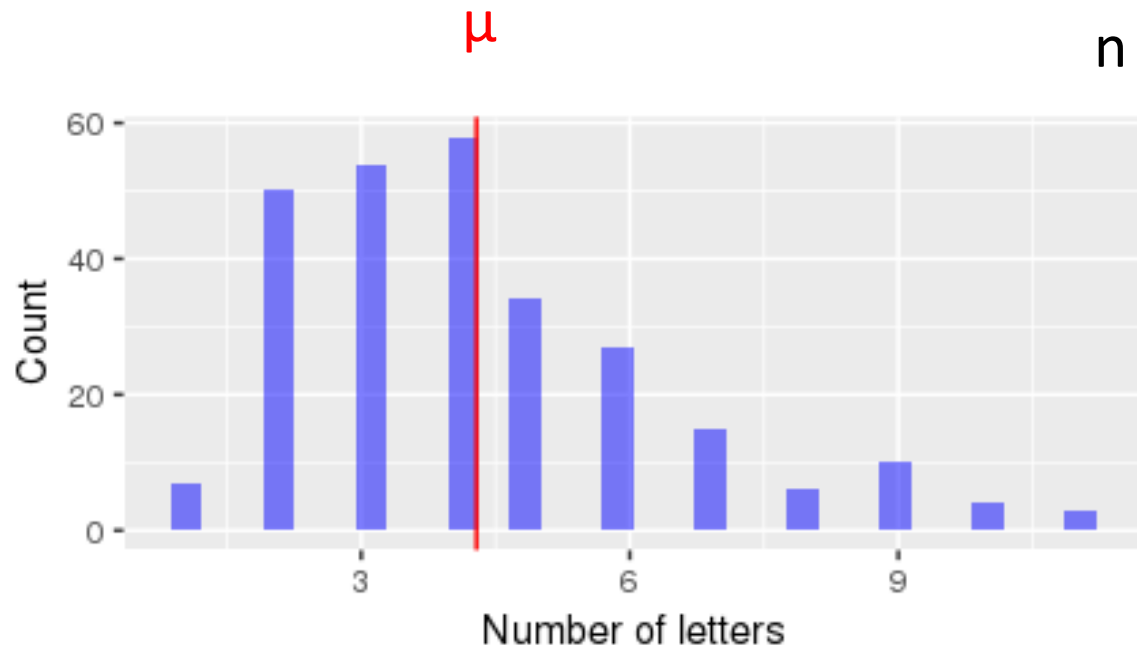
Distribution
of $\hat{p}$'s



Sampling distribution!



Gettysburg address word length sampling distribution



10, 3, 3, 3, 4,
3, 2, 6, 10, 5

$$\bar{x} = 5$$

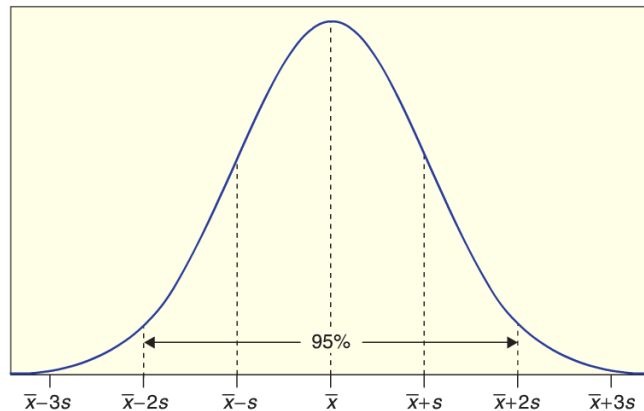
2, 6, 2, 6, 6,
2, 5, 3, 2, 9

$$\bar{x} = 4.3$$

3, 9, 3, 4, 4,
3, 6, 6, 2, 2

$$\bar{x} = 4.2$$

Distribution
of $\bar{x}$'s



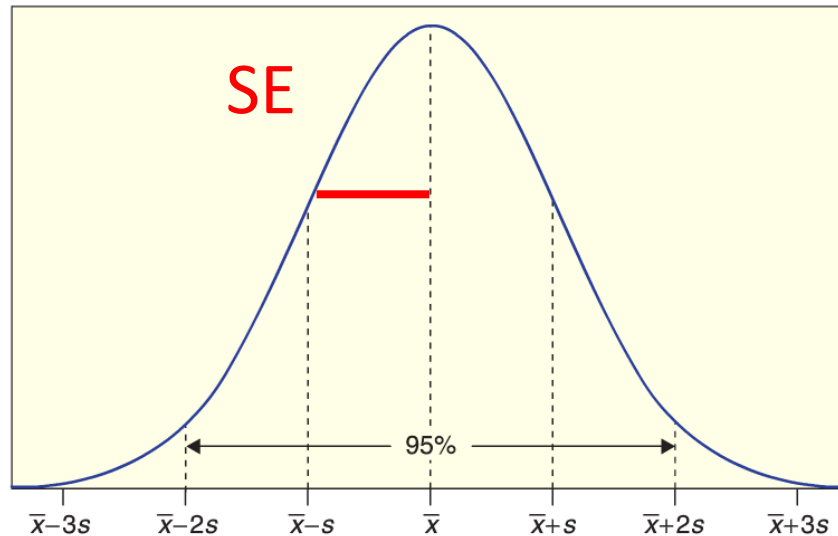
Sampling distribution!

[Gettysburg sampling distribution app](#)

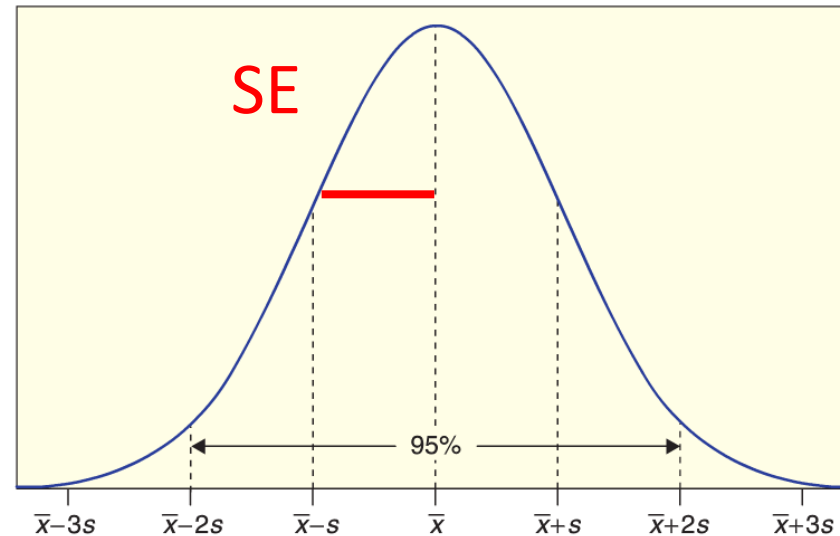
The standard error

The **standard error** of a statistic, denoted SE, is the standard deviation of the sample statistic

- i.e., SE is the standard deviation of the *sampling distribution*



What does the size of a standard error tell us?



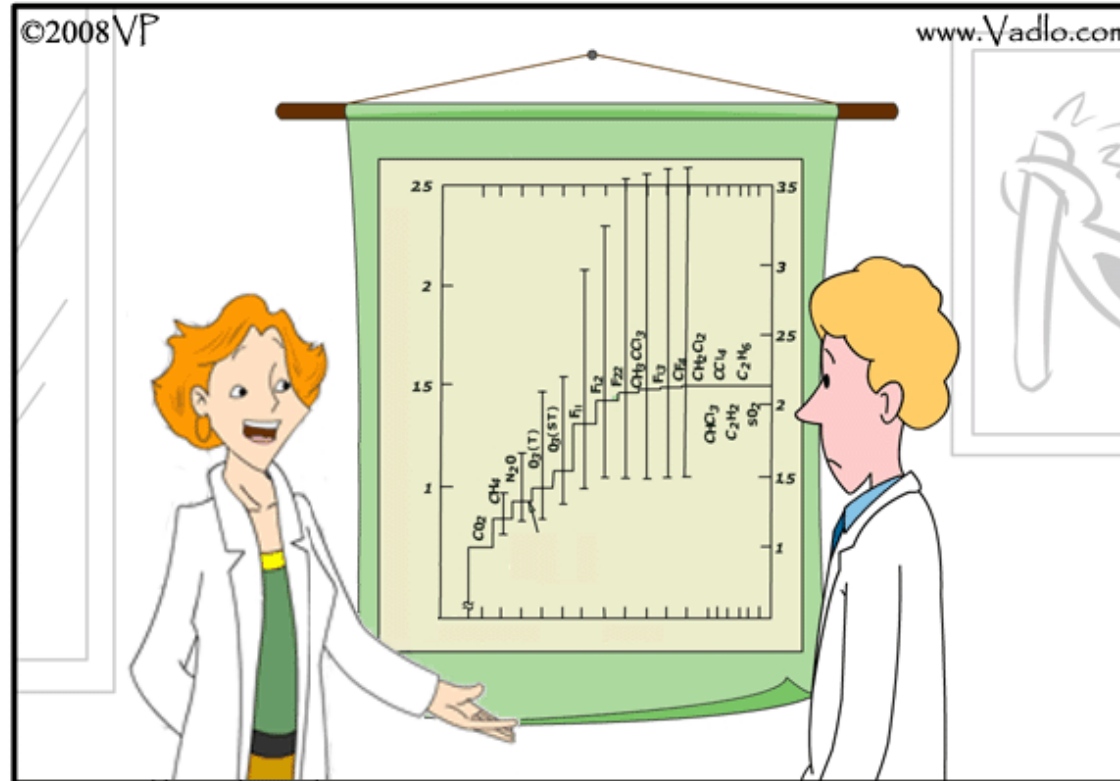
Q: If we have a large SE, would we believe a given statistic is a good estimate for the parameter?

- E.g., would we believe a particular $\bar{x}$ is a good estimate for μ ?

A: A large SE means our statistic (point estimate) could be far from the parameter

- E.g., $\bar{x}$ could be far from μ

Questions?



Did you really have to show the error bars?

Sampling distributions in R!

Let's create a sampling distribution in R

Load the SDS1000 library to make all SDS1000 functions available

```
library(SDS1000)
```

Get the Gettysburg population data

```
load("gettysburg.Rda")
```

```
word_lengths <- gettysburg$num_letters # lengths of the 268 words
```

Let's create a sampling distribution in R

We can use the `sample(data_vec, n)` to get a sample of length n:

```
curr_sample <- sample(word_lengths, 10)
```

Q: How can we get $\bar{x}$ from this sample in R?

```
mean(curr_sample)
```

Q: How could we get a full sampling distribution?

- A: Repeat this many times to get an approximation of the sampling distribution
- If we store the $\bar{x}$'s in a vector, we can then plot the sampling distribution as a histogram

The do_it() function

The `do_it()` function (from the SDS1000 package) repeats a piece of code many times

- It returns a vector with the values created each time the code is repeated

```
do_it(100) * {
```

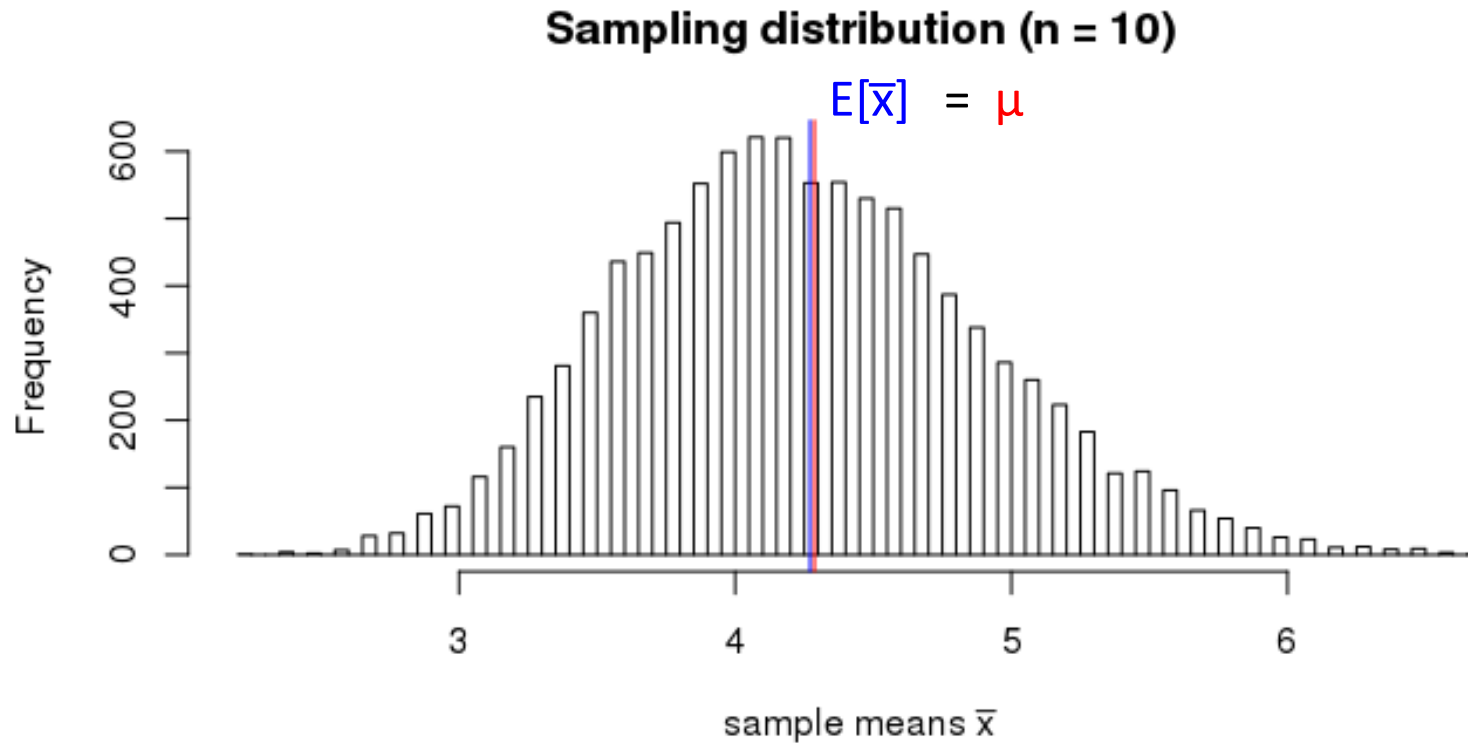
```
  2 + 3
```

```
}
```

Let's create a sampling distribution in R

```
sampling_dist <- do_it(10000) * {  
  
    curr_sample <- sample(word_lengths, 10)  
    mean(curr_sample)  
  
}  
  
hist(sampling_dist)
```

Sampling distribution in R



```
mean(sampling_dist)
```

```
mean(word_lengths) # these are the same, so no bias
```

Changing the sample size n

What happens to the sampling distribution as we change n ?

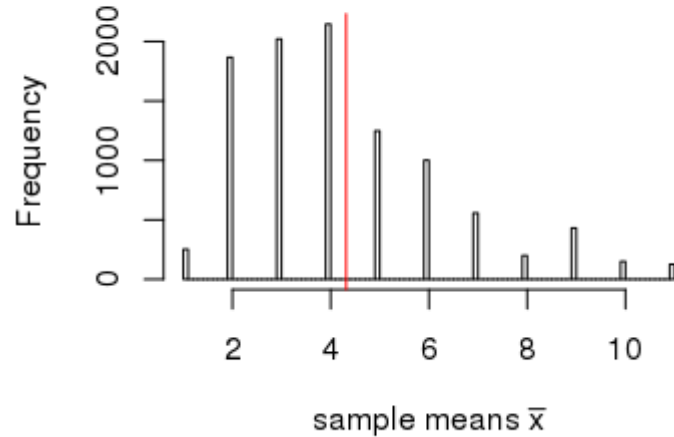
- Experiment for $n = 1, 5, 10, 20$

```
sampling_dist <- do_it(10000) * {  
    curr_sample <- sample(word_lengths, 20)  
    mean(curr_sample)  
}
```

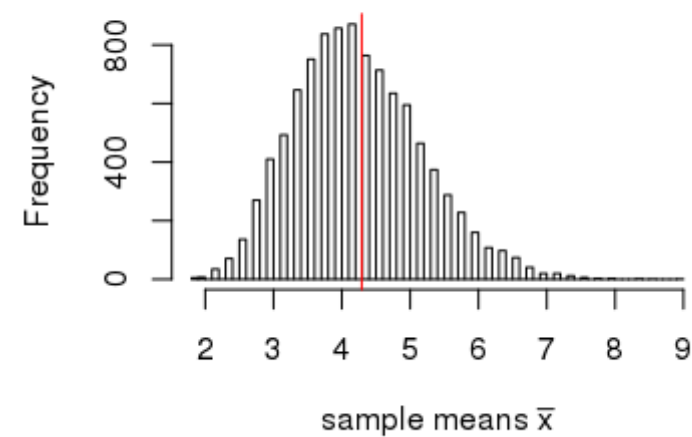
```
hist(sample_means, breaks = 100)
```

[Gettysburg sampling distribution app](#)

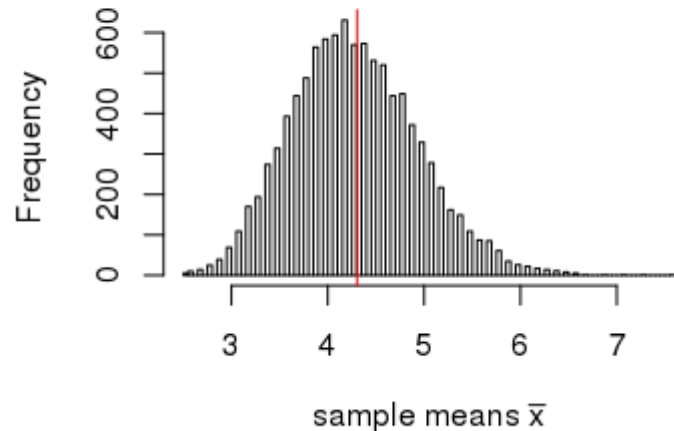
Sampling distribution ($n = 1$)



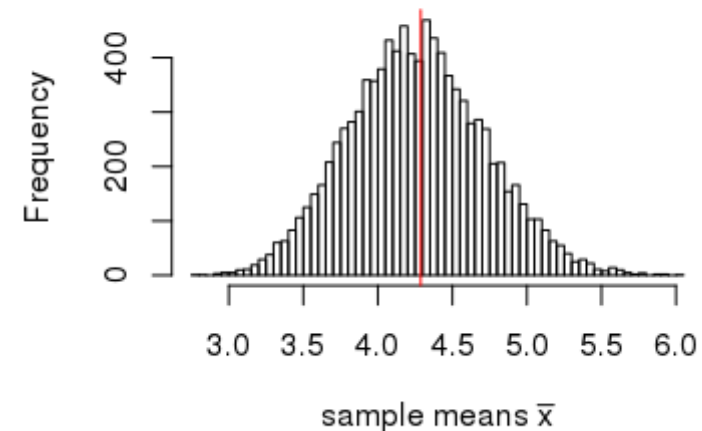
Sampling distribution ($n = 5$)



Sampling distribution ($n = 10$)



Sampling distribution ($n = 20$)



x-axis range 9 vs. 6

As the sample size n increases

1. The sampling distribution becomes more like a normal distribution
2. The sampling distribution points ($\bar{x}$'s) become more concentrated around the mean $E[\bar{x}] = \mu$